

Operational Limits of Ontological Innovation in LLMs: Evidence from Controlled Sensory-Modality Generation

Jarno Matarmaa^{1*}

^{1*}Institute of Radio Electronics and Information Technologies, Ural Federal University, Yekaterinburg, Russia.

Corresponding author’s email: iarnoolavi.matarmaa@urfu.ru;

Abstract

Contemporary large language models generate text that can appear conceptually novel, raising the question of whether outputs reflect genuine ontological innovation—the introduction of concepts that cannot be expressed as recombinations or transformations of training-derived primitives—or sophisticated recombination within a fixed representational manifold. We test this via a controlled experiment in an eight-modality sensory ontology. Seven language models ($N = 168$ proposals; 24 runs each) were prompted to propose a genuinely new ninth modality. Outputs were evaluated with four convergent methods: literature traceability ($\tau = \mathbf{0.40}$), structural decomposition, embedding-space convex-hull analysis, and continuous novelty scoring. Results show a strict novelty rate of $\hat{p}_{\text{novel}} = \mathbf{0.000}$ (95% CI: [0.000, 0.000]), traceability $\hat{p}_{\text{traceable}} = \mathbf{0.982}$, and 100% convex-hull membership under Sentence-BERT embeddings (3 PCA components). Structural decomposition revealed three dominant failure modes: range extension (20.2%), hybrid combination (44.6%), and functional recombination (17.9%). Cross-model effects were moderate ($\eta^2 = \mathbf{0.326}$), but all models stayed within the same low-novelty regime, supporting a constrained-manifold interpretation. Under the *No New Basis Theorem*, gradient-descent learning over fixed vocabularies cannot expand representational dimensionality beyond training data. Current architectures therefore appear bounded by training-derived conceptual closures.

Keywords: ontological innovation, computational creativity, large language models, conceptual spaces, traceability analysis, representational constraints

1 Introduction

The capacity to generate fundamentally new concepts—not merely recombinations of existing ones—has long been considered a hallmark of human intelligence and a defining test of genuine creativity. As large language models (LLMs) achieve impressive performance on a broadening range of generative tasks, a pressing question has emerged: do these systems exhibit *ontological innovation*, generating ideas that lie outside the conceptual closure of their training environment, or do they operate as extremely capable recombinators constrained to a fixed representational manifold?

This question has both philosophical and practical stakes. Philosophically, it bears on the computational theory of mind, the nature of creativity, and whether algorithmic systems can in principle exceed the representational resources with which they were trained. Practically, it determines what roles LLMs can occupy in scientific discovery, design, and other domains where genuinely new concepts—not just novel arrangements of existing ones—are required [1].

The challenge is that outputs from modern LLMs frequently *appear* novel. Proposals that introduce unfamiliar terminology, combine concepts across disciplinary boundaries, or describe mechanisms not explicitly present in any single training document can pass casual inspection as genuinely creative [2, 3]. Evaluating whether such outputs constitute true conceptual innovation requires a controlled experimental setup and rigorous multi-method analysis rather than surface-level assessment.

In this paper we report a focused empirical study designed to provide exactly this kind of evaluation. We use a multimodal sensory ontology—a well-defined, eight-modality conceptual space representing biological and extended machine sensing—as the experimental universe. This setup provides precise control: we know exactly which concepts exist in the training context, enabling systematic investigation of whether AI-generated proposals lie outside the defined space. Seven contemporary LLMs were prompted to propose a *ninth* sensory modality, genuinely beyond any of the eight. We evaluate their proposals through four convergent methods that progressively sharpen the diagnostic picture.

Our main contributions are:

1. A controlled experimental paradigm for operationalising ontological innovation, grounded in a well-defined multi-modal sensory ontology that functions as an explicit training universe.
2. Multi-method evaluation combining literature traceability with calibrated thresholds, structural decomposition, embedding-space convex-hull analysis, and continuous novelty scoring, providing convergent evidence without reliance on subjective expert judgment.
3. Systematic empirical evidence that seven state-of-the-art LLMs produce zero strictly novel proposals across 168 evaluated outputs, with near-saturated traceability to known conceptual material.
4. Formal grounding through the *No New Basis Theorem*, relating observed empirical constraints to architectural properties of gradient-descent-trained neural networks.

The paper is organized as follows. Section 2 reviews related work on computational creativity and representational limits. Section 3 describes the dataset, task, models,

and evaluation procedures. Section 4 presents the results across all evaluation methods. Section 5 interprets the findings, and Section 6 concludes.

2 Background and Related Work

2.1 Boden’s Creativity Taxonomy

The dominant framework for analysing machine creativity is Margaret Boden’s tripartite taxonomy [4], which distinguishes outputs based on their relationship to the conceptual space from which they arise. *Combinational creativity* produces novel combinations of familiar ideas—poetic metaphors, analogies, conceptual blends. *Exploratory creativity* systematically searches well-defined conceptual spaces to find previously unvisited regions. *Transformational creativity*, the most radical category, modifies the conceptual space itself by relaxing or changing its defining constraints.

Research on AI systems has demonstrated substantial competence at combinational and exploratory creativity [5, 6]. Systems such as Copycat, JAPE, and AARON can generate outputs that human evaluators judge as genuinely surprising and apt within their respective domains. Robust transformational creativity, however, remains rare: most documented examples involve superficial parameter modifications rather than fundamental reconceptualisation [4].

Wiggins [2] introduced a formal framework for evaluating creativity claims and identified a “framing” problem: systems appear to transform their conceptual space while actually operating within an implicitly broader space defined by human designers. This critique applies with particular force to contemporary LLMs, whose training corpora implicitly define a vast but still finite conceptual universe.

2.2 Our Extension: Ontological Creativity

We argue that a fourth category, which we term *ontological creativity*, is needed to capture the most radical form of conceptual innovation. Transformational creativity modifies operators Ω or relations $\mathcal{E}$ within a conceptual space $\mathcal{C} = (V, \mathcal{E}, \Phi, \Omega)$; ontological creativity introduces new *primitive* concepts $v' \in V'$ where $V' \not\subseteq V$ and critically V' cannot be expressed as $\Omega(V)$ for any existing operators.

This distinction maps onto Ryle’s philosophical notion of category mistakes [7]: ontological creativity involves recognising that existing categories are insufficient and proposing new ones that cannot be defined by recombining or transforming current primitives. Einstein’s reconceptualisation of absolute simultaneity as a category error, leading to a revised ontology of relativistic spacetime, exemplifies this form of creativity. Generating a genuinely new sensory modality would be analogous: not an extension or hybridisation of existing ones, but a proposal for a sensing type with a physical basis and information content that cannot be expressed as a composition of the existing eight.

2.3 Formal Representational Constraints

The computational theory of mind [8, 9] and its critics [10–12] have long debated whether algorithmic systems face principled limits on what they can represent and

understand. Recent work in mechanistic interpretability provides empirical traction on this question by revealing that generation in transformer-based LLMs proceeds through retrieval and interpolation among training examples rather than through any inference mechanism that could introduce entirely new primitives [13, 14]. The transformer architecture [15] encodes all knowledge implicitly in weights learned from distributional co-occurrence, which grounds the output space in the training vocabulary of concepts. Gärdenfors’s conceptual-spaces formalism [16] provides complementary geometric intuition: a learned concept lives as a region in a quality space, and generating a new concept outside all existing regions requires an external basis vector not present in the training data.

Claim (No New Basis Theorem under fixed-basis assumptions). For a neural architecture $\mathcal{A}$ operating over representation space $\mathbb{R}^d$, with generative process $G: \mathbb{R}^d \rightarrow \mathbb{R}^d$ learned from training data $\mathcal{D} = \{x_1, \dots, x_N\}$, generated outputs $x' = G(z)$ remain within a manifold $\mathcal{M}'$ that is a continuous deformation of $\mathcal{M} = \text{span}(\{x_1, \dots, x_N\})$, such that $\dim(\mathcal{M}') \leq \dim(\mathcal{M})$.

Assumptions. (i) fixed representational basis and vocabulary during inference, (ii) gradient-based continuous learning dynamics, and (iii) no external mechanism that appends new primitive basis vectors at inference time.

Argument (informal proof sketch). First, under continuous gradient updates, learned mappings deform regions of the training-induced manifold rather than creating independent coordinate axes. Second, generative decoding composes these learned maps and therefore remains constrained by the same basis. Third, introducing ontological primitives outside the learned span would require explicit basis augmentation (new symbols, new sensors, or external memory with independent representational coordinates), which is excluded by assumptions (i)–(iii).

Consequence. Ontological novelty, defined as basis expansion rather than recombination, cannot be produced by the internal generation mechanism alone under these assumptions.

This theorem explains why even highly capable models that produce surprising combinations remain formally constrained to the conceptual universe their training data defines. Our empirical study provides a controlled test of whether this theoretical constraint manifests behaviorally.

2.4 Evaluation Challenges in Computational Creativity

A persistent methodological challenge in computational creativity research is evaluation [2, 3]. Human expert judgments are subjective, difficult to reproduce, and sensitive to framing. Recent studies show that LLMs can achieve human-level performance on divergent-thinking tasks such as the Alternative Uses Test [17], but scoring quality depends strongly on the evaluation metric [18]. Recent 2024–2026 work further confirms metric sensitivity and proposes newer creativity-evaluation protocols for LLMs, including human-comparative, temperature-sensitivity, and model-guided creativity-judging paradigms [19–25]. Our multi-method approach—traceability search, structural decomposition, embedding-space geometry, and continuous scoring—provides

Table 1 Eight-modality sensory ontology used as the experimental universe. Each modality is defined by its physical basis, representational dimensionality, and functional role.

Modality	Physical basis	Functional role	Dim.
Visual	EM spectrum 300–1000 nm (5 nm steps)	Scene and object recognition	140
Auditory	Pressure waves 1 Hz–100 kHz (128-bin FFT)	Sound source localisation	128
Tactile	Mechanical vibrations 5–800 Hz	Surface texture and pressure	32
Olfactory	Chemical categories (10 classes)	Substance identification	10
Gustatory	Taste dimensions (sweet/sour/salty/bitter/umami)	Ingestion guidance	5
Proprioceptive	20 joint angles + 10 muscle tensions	Body-state estimation	30
Vestibular	3-axis acceleration + 3-axis angular velocity	Balance and motion	6
Interoceptive	HR, respiration, skin conductance, temperature, hunger, thirst, fatigue, pain	Homeostatic regulation	8
<i>Total feature dimensionality per timestep</i>			359

Note on Dim. Dimensionality values represent the number of independently controllable encoding parameters for each modality stream (e.g., spectral bins, frequency bins, category classes, kinematic channels, and homeostatic channels), not biological receptor counts.

convergent automated evidence that can be replicated and extended. Calibrated traceability thresholds are used as empirically justified operating points for interpretation rather than as unconditional binary boundaries.

3 Materials and Methods

3.1 The Multimodal Sensory Ontology

The experimental universe consists of eight sensory modalities, each defined by a physical basis, an information channel, and a functional role in an agent’s interaction with its environment (Table 1). Modalities span biological human senses and extended machine-sensing capabilities:

The full dataset contains 5,000 training, 1,000 validation, and 1,000 test examples, each representing a 60-second episode at 1 Hz sampling (21,540 data points per example) drawn from seven everyday scenarios (walking, running, sitting, eating, drinking, listening, viewing). An explicit machine-readable ontology layer documents the physical basis, information content, and functional role of each modality, enabling automated evaluation.

3.2 Task Specification and Prompt Design

A task specification (prompt contract) is a structured set of constraints that every prompt instantiation must satisfy, ensuring comparability across models and runs.

Each model was asked to *propose a ninth sensory modality that is genuinely new and not reducible to combinations, extensions, or transformations of the existing eight.*

Proposals were required to specify: (1) the physical phenomenon to be sensed, (2) the information channel and encoding, and (3) the functional difference from existing modalities. The complete ontology table (Table 1) was provided in structured tabular format alongside a natural-language description of each modality.

Prompts were designed to be task-neutral—they did not suggest what kinds of answers would or would not qualify as novel, to avoid anchoring models toward specific response types. Twenty-four independent runs were collected from each of seven models, yielding $N = 7 \times 24 = 168$ proposals in total.

3.3 Models and Implementation

Seven state-of-the-art models were evaluated: GPT-5.2 [26], Claude-3.7-Sonnet, Gemini-3.1-Pro-Preview [27], LLaMA-3.3-70B-Instruct [28], DeepSeek-v3.2 [29, 30], Mistral-Large [31], and Perplexity Sonar Pro, accessed via the OpenRouter, Perplexity, and Mistral APIs. All runs used temperature $T = 0.7$; the Perplexity API does not support temperature adjustment and was operated at provider defaults. System prompts were crafted to neutrally present the task without introducing bias toward particular answer types.

3.4 Evaluation Protocol

Literature traceability.

Each proposal was matched against a curated scholarly corpus of 125 documents covering sensory biology, robotics sensing, and computational neuroscience, using cosine similarity over Sentence-BERT embeddings [32]. A proposal is classified as *traceable* if its best-match similarity exceeds threshold τ .

The corpus was assembled from OpenAlex [33] and Crossref API retrieval using domain seed queries, then deduplicated and length-filtered before embedding. OpenAlex and Crossref were used as discovery/indexing layers for scholarly records, while traceability labels were assigned by semantic similarity on cleaned text rather than by keyword overlap alone.

The corpus scope is deliberately academic-mainstream. In this study, traceability is defined with respect to scientific literature in those three domains rather than science-fiction, patent, or broader philosophical corpora.

Threshold calibration followed an explicit interpretive-design criterion: the operating point should avoid degenerate label collapse (e.g., near-universal traceability or near-universal non-traceability), because such regimes are weakly informative for comparative inference. Three candidate operating points ($\tau \in \{0.30, 0.40, 0.50\}$) were therefore evaluated on the full response–corpus similarity distribution to quantify direction and rate of change in traceability as strictness increases. At $\tau = 0.30$, the traceability rate was 0.982, indicating a near-saturated regime with limited discriminatory power. At $\tau = 0.50$, only 64.9% of proposals were traceable, indicating substantial pruning. The intermediate value $\tau = 0.40$ was used for the main text analysis as a calibrated reference point for inferential inspection. It is interpreted as an operational indicator of current position in the similarity distribution, not as an absolute boundary that unconditionally separates traceable from non-traceable outputs. To address

Table 2 Sensitivity of core outcomes to traceability threshold τ .

τ	Traceability rate	Strict novelty rate	Mean continuous novelty
0.30	0.982	0.000	0.059
0.40	0.982	0.000	0.059
0.50	0.649	0.000	low/stable

threshold-dependence concerns, Table 2 reports key outcomes at all three thresholds and shows that strict novelty remains zero.

Structural decomposition.

Each proposal was analysed for whether its key claims can be expressed using only the eight training primitives through: (i) range extension (preserving the sensing type while shifting its physical parameter range), (ii) hybrid combination (combining features from two or more existing modalities), or (iii) functional recombination (repurposing an existing sensing mechanism for a different functional role). Proposals not fitting any category were labelled *structurally novel*.

Embedding-space convex-hull analysis.

Sentence-BERT embeddings were computed for each proposal and for the eight training-modality descriptions. Principal component analysis reduced both sets to three components; convex-hull membership was computed in 3D using half-space equations with numerical tolerance. A proposal classified as *outside* the hull would occupy a semantically isolated region suggesting basis expansion beyond the training ontology.

Continuous novelty scoring.

A continuous novelty score $s \in [0, 1]$ was computed for each proposal by combining structural novelty sub-scores: (i) absence from structural decomposition categories, (ii) inverse of maximum cosine similarity to training modalities, and (iii) geometric distance from the convex-hull boundary. Proposals with $s \geq 0.5$ were classified as strictly novel under this operational threshold.

4 Results

4.1 Primary Outcomes

Before interpreting the null novelty result, semantic coherence was assessed informally across all 168 proposals. Every proposal described a physically plausible sensing mechanism with an identifiable input channel and functional role; none were semantically degenerate or nonsensical. This matters because incoherent out-of-distribution text would not count as ontological innovation.

The aggregate strict novelty rate was $\hat{p}_{\text{novel}} = 0.000$ (95% CI: [0.000, 0.000]; $N = 168$). No proposal satisfied the strict novelty criterion $s \geq 0.5$. The mean continuous

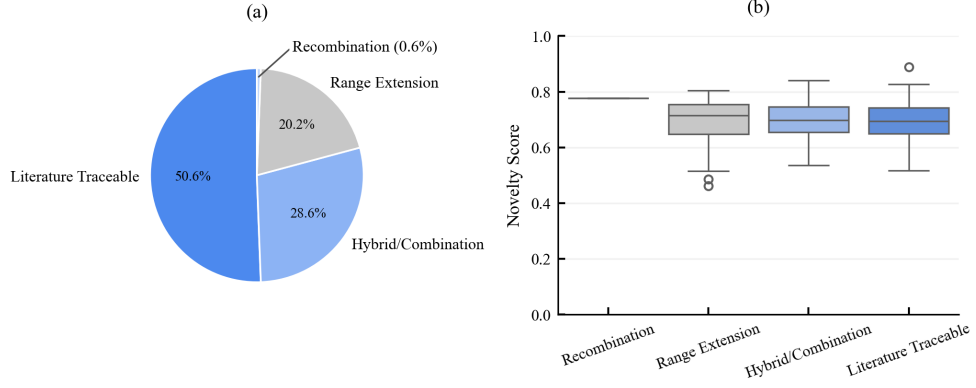


Fig. 1 Classification summary ($N = 168$ proposals, seven models). (a) Distribution of classification outcomes; the distribution is dominated by traceable categories consistent with low operational ontological novelty under calibrated thresholds. (b) Novelty score distributions by classification category shown as box plots.

novelty score was $\hat{\mu}_{\text{continuous}} = 0.059$ (SD = 0.041), indicating weak novelty signal without threshold-level novelty events across any model or run.

Under the calibrated traceability threshold $\tau = 0.40$, the traceability rate was $\hat{p}_{\text{traceable}} = 0.982$ (95% CI: [0.962, 1.000]); 165 of 168 proposals had a best-match cosine similarity above threshold, with a mean best-match score of 0.528 across all proposals. Of the 165 traceable proposals, all 165 were matched to sources from the strong-evidence tier of the corpus. Consistent with Table 2, this threshold should be read as an operational reference value: the directional pattern remains stable at lower strictness and transitions toward stronger pruning at higher strictness.

Figure 1 presents the dominant classification split: traceable/non-novel outputs are overwhelmingly prevalent across all models.

4.2 Structural Decomposition

Of the 168 proposals, 88 (52.4%) were structurally decomposable into training-ontology primitives. The detailed outcome groups are summarized in Table 3. Three failure modes account for the decomposable proposals:

- **Range extension** (34 cases, 20.2%): proposals that extend the physical parameter range of an existing modality (e.g., far-infrared thermal sensing as an extension of visual EM-band coverage, or infrasound-extended auditory detection).
- **Hybrid combination** (75 cases, 44.6%): proposals combining features from two or more existing modalities (e.g., combined tactile-proprioceptive “haptic flow” sensing, or olfactory-gustatory integration).
- **Functional recombination** (30 cases, 17.9%): proposals repurposing an existing sensing mechanism for a different functional role (e.g., using interoceptive signals for external-environment monitoring rather than homeostatic regulation).

Table 3 Structural decomposition outcome groups.

Group	Count	Interpretation
Fully structurally decomposable	88 (52.4%)	Mapped to named patterns (possibly overlapping)
Structurally heterogeneous, semantically proximate	80 (47.6%)	Not cleanly in one pattern, but traceable

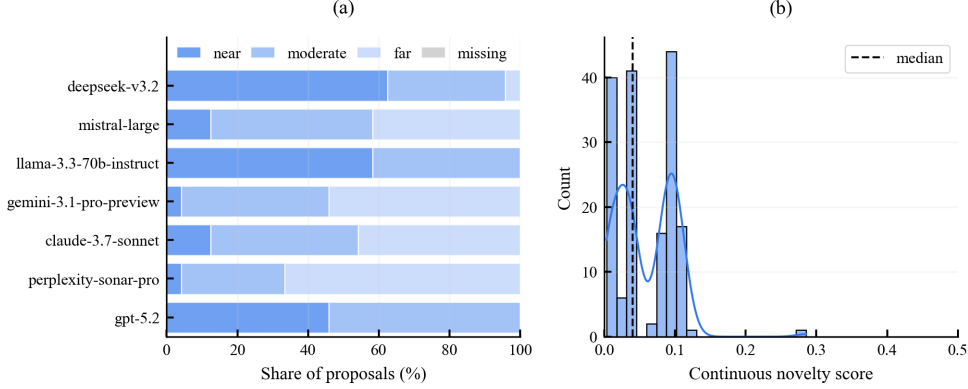


Fig. 2 Novelty-frontier band assignments and score distribution. (a) Band assignment by model: responses are concentrated in low-novelty and frontier-proximal bands across all seven architectures, indicating architecture-independent constraint patterns rather than isolated model failures. (b) Overall continuous novelty score distribution with KDE overlay; dashed line marks the median.

The three category counts overlap: 51 proposals exhibited more than one pattern simultaneously, so the non-overlapping count of fully decomposable proposals is 88. The remaining 80 proposals (47.6%) were analyzed but did not fit cleanly into one named structural pattern; these are better interpreted as a structurally heterogeneous yet semantically proximate group.

Figure 2 shows novelty-frontier band assignments by model, confirming that the low-novelty/high-traceability pattern holds architecture-wide rather than being driven by a single underperforming model. Figure 3 further illustrates that most proposals fail multiple frontier criteria simultaneously, supporting a coupled-constraint interpretation.

4.3 Embedding-Space Geometry

All 168 proposals (100%) were classified as inside the 3D convex hull of training-modality embeddings. Proposal markers cluster in the interior region of the training manifold; no proposal occupies a semantically isolated region that would suggest basis expansion beyond the training ontology.

The mean cosine similarity between each proposal and its closest training modality was $\bar{d} = 0.305$ (SD = 0.078), indicating substantial semantic overlap with training

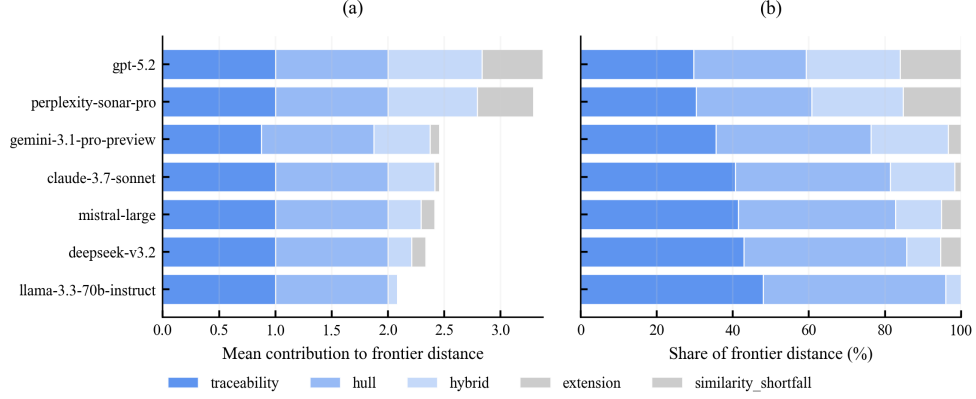


Fig. 3 Novelty-frontier criterion decomposition by model. (a) Absolute mean contribution of each criterion to overall frontier distance. (b) Relative share (%) of frontier distance per criterion. Both panels confirm that most responses violate multiple criteria simultaneously, supporting a coupled-constraint interpretation of low ontological novelty.

concepts. Figure 4 presents the embedding-space analysis with proposals visualised in the first two principal components and the 3D hull boundary shown as a visual aid.

Figure 5 provides the aggregated cosine-similarity heatmap of proposed modalities against the eight training modalities, reaching the same conclusion from a complementary metric: every proposal records elevated similarity to one or more training modalities, with no proposal showing uniformly low similarity across all eight.

4.4 Cross-Model Comparison

The model effect on the primary novelty metric was moderate: one-way ANOVA yielded $F(6, 161) = 12.98$, $p < 0.001$, $\eta^2 = 0.326$, indicating non-trivial between-model variance in raw scores. Nonetheless, all seven models remained within the same low-novelty/high-traceability regime; no model produced proposals classified as strictly novel, and no model achieved a mean continuous novelty score exceeding $\mu = 0.12$.

Post-hoc Tukey HSD comparisons are summarized in Table 4. The full pairwise pattern is selective rather than global: GPT-5.2 scores significantly lower than Claude-3.7-Sonnet, DeepSeek-v3.2, Gemini-3.1-Pro-Preview, LLaMA-3.3-70B-Instruct, and Mistral-Large; Perplexity Sonar Pro scores significantly lower than Claude-3.7-Sonnet, Gemini-3.1-Pro-Preview, LLaMA-3.3-70B-Instruct, and Mistral-Large. The remaining pairwise differences are not significant after multiplicity correction.

Figure 6 shows that between-model variation takes the form of different *preferences among known canonical families* rather than emergence of structurally new families: some models more frequently produced range-extension proposals while others favoured hybrid combinations, but neither pattern represents departure from the training-derived manifold.

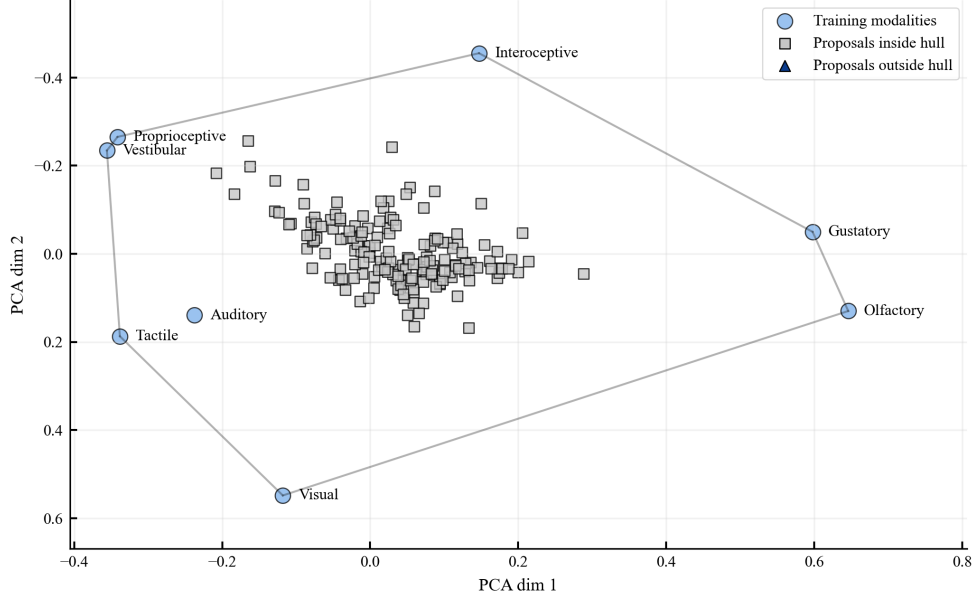


Fig. 4 Embedding-space analysis for proposals and training modalities. Convex-hull membership is computed in PCA-reduced 3D space; the 2D boundary shown is illustrative only. All 168 proposal markers cluster near the interior of the training manifold—none escape to a semantically isolated region.

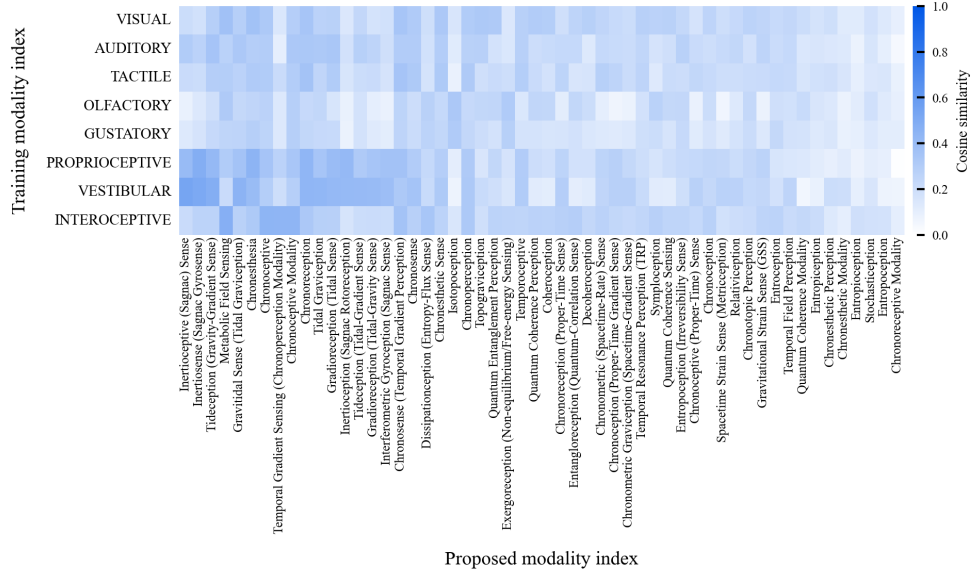


Fig. 5 Cosine similarity of proposed modalities to the eight training modalities. Columns: aggregated unique proposed modality index (1–54); rows: training modality index (1–8); The concentration of cosine-similarity values is consistent with the traceability result: no aggregated proposal appears uniformly dissimilar to all eight training modalities.

Table 4 Tukey HSD post-hoc comparisons for continuous novelty score. Significant contrasts are listed individually; non-significant contrasts are aggregated. The ‘Significant’ column indicates whether the Tukey HSD multiplicity-adjusted p-value falls below the prespecified threshold of $\alpha = 0.05$.

Comparison	Adjusted p-value	Significant
Claude-3.7-Sonnet vs. Perplexity Sonar Pro	0.0076	Yes
Mistral-Large vs. Perplexity Sonar Pro	0.0043	Yes
Gemini-3.1-Pro-Preview vs. Perplexity Sonar Pro	0.0005	Yes
LLaMA-3.3-70B-Instruct vs. Perplexity Sonar Pro	0.0001	Yes
Claude-3.7-Sonnet vs. GPT-5.2	0.0000	Yes
DeepSeek-v3.2 vs. GPT-5.2	0.0000	Yes
Gemini-3.1-Pro-Preview vs. GPT-5.2	0.0000	Yes
GPT-5.2 vs. LLaMA-3.3-70B-Instruct	0.0000	Yes
GPT-5.2 vs. Mistral-Large	0.0000	Yes
All remaining pairwise contrasts (n = 12)	≥ 0.05	No

Figure 7 presents continuous novelty score distributions per model as a ridgeline plot. Distributional overlap is high across all seven models, and peaks consistently remain in low-to-moderate ranges, indicating that model-specific differences in architecture and scale do not qualitatively alter the constraint pattern.

4.5 Summary of Results

The four evaluation methods converge on a coherent picture: across 168 proposals from seven state-of-the-art LLMs, zero proposals satisfy the strict novelty criterion, 98.2% are traceable to the training corpus at the calibrated reference threshold, and every proposal lies inside the training-derived convex hull in embedding space. The structural decomposition reveals that the proposals explore recognisable combinatorial and extension patterns rather than introducing new ontological primitives. Cross-model effects influence *degree* of constraint rather than its qualitative nature.

5 Discussion

5.1 Constrained Manifold Interpretation

The consistent null result across models, evaluation methods, and threshold sensitivity checks supports a *constrained-manifold* interpretation: LLMs generate outputs within a representational manifold shaped by training data, and this manifold does not expand during inference regardless of prompt framing, model scale, or generation temperature.

This interpretation aligns with the No New Basis Theorem stated in Section 2. The theorem establishes that generative processes learned via gradient descent on continuous loss functions cannot increase the intrinsic dimensionality of the output manifold beyond the span of training data. Our empirical results are consistent with this prediction: even the continuous novelty scores, designed to reward partial innovation, show no threshold-level events and a mean value close to zero.

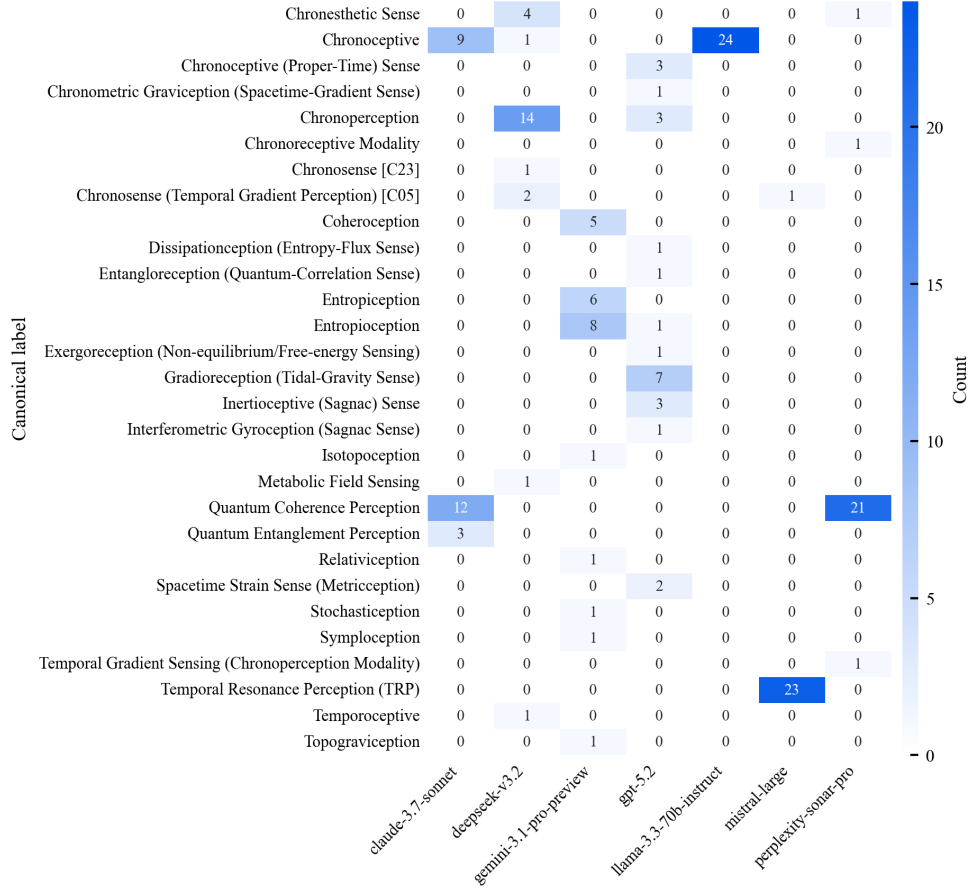


Fig. 6 Canonical proposal clusters by model. Between-model variation manifests primarily as preference differences among existing canonical families (range extension, hybrid combination, functional recombination) rather than as emergence of qualitatively new ontological families. Note: [Cxx] marks distinct raw clusters when canonical labels share the same base name.

Gödel’s incompleteness results [9] and Turing’s halting-problem proof [8] provide deeper formal context. If recognising the inadequacy of one’s own conceptual framework requires a form of self-reference analogous to constructing a Gödel sentence, then this capacity may be unavailable to computational systems operating within fixed formal specifications. Our empirical evidence is consistent with this theoretical prediction, though it does not constitute a formal proof that such incapacity is necessary rather than contingent.

5.2 Relationship to the Creativity Taxonomy

Within Boden’s taxonomy [4], the structural decomposition results place nearly all proposals firmly in the combinational and exploratory categories: proposals are either novel combinations of existing primitives (hybrid combinations) or new regions of

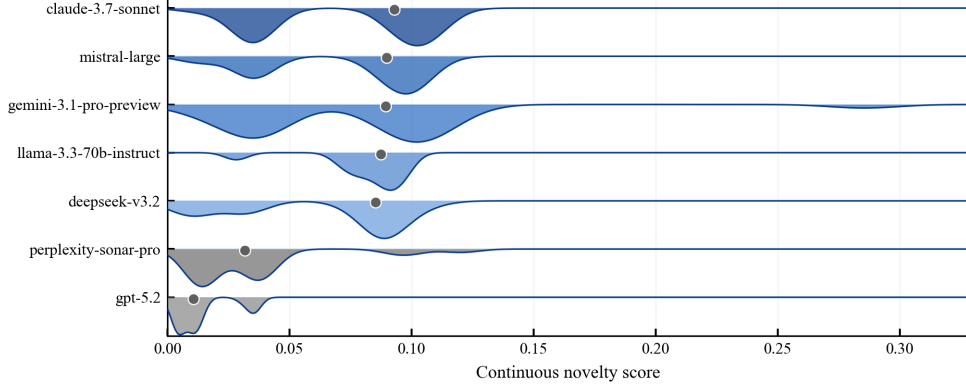


Fig. 7 Continuous novelty score by model (ridgeline plot). High distributional overlap and consistently low-to-moderate peaks indicate that model scale and architecture do not qualitatively alter the constrained-manifold pattern.

existing modality spaces (range extensions). A small fraction involve functional recombination, which could be viewed as approaching transformational creativity, but the transformations remain within the ontological type system defined by the training modalities rather than introducing new types.

The absence of any proposal in the ontological creativity category supports Wiggins’s [2] observation that apparent transformational creativity in AI systems frequently reflects an implicitly broader framing space defined by human designers. In our setup the framing space is made explicit through the eight-modality ontology, enabling a precise test: proposals must introduce a sensing type genuinely outside this space. None do.

5.3 Alternative Explanations

Several alternative explanations deserve consideration:

Scale.

All seven models span a wide range of scales, from the 70B-parameter LLaMA-3.3-70B to large-scale commercial systems such as GPT-5.2 and Claude-3.7-Sonnet. The $\eta^2 = 0.326$ model effect indicates that scale influences raw novelty scores, but no model achieves strict novelty. This suggests the constraint is not a contingent result of insufficient scale within the tested range.

Prompt sensitivity.

The present study used a fixed prompt design with temperature ablation limited to $T = 0.7$. We cannot rule out that alternative prompt formulations might yield marginally higher novelty scores, though they would need to change not just lexical generation patterns but the geometric structure of the output manifold. Future work should include systematic prompt-paraphrase grids to test robustness.

Evaluation sensitivity.

Conservative thresholds mean we may underestimate AI creativity by classifying some proposals as traceable that an expert would consider novel. Conversely, overly strict thresholds can collapse traceability labels and reduce interpretive value. We therefore treat thresholding as a calibrated measurement device for locating current position and movement within the response–corpus similarity landscape, not as an unconditional criterion of ontological status. However, the embedding-space analysis is threshold-free, and 100% hull membership provides threshold-independent evidence of constrained generation.

Recombination as the norm.

One counterargument holds that human ontological creativity may also, on careful analysis, involve recombination of prior concepts in ways we fail to recognise due to incomplete historical knowledge of our own cognitive processes. We cannot definitively rule this out, but the structural patterns here—range extension, hybrid combination, functional recombination—are qualitatively different from the kind of reconceptualisation exemplified by the introduction of non-Euclidean geometry or special relativity, even granting that such innovations drew on prior materials [1, 34].

5.4 Implications

The results have direct implications for deploying LLMs in discovery-oriented tasks. Within Kuhnian normal science [1]—systematic exploration of hypothesis space within an established paradigm—LLMs offer substantial value through exhaustive combinatorial coverage and perfect recall. The success of AlphaFold in predicting protein structures [35] illustrates how AI can achieve transformative impact within a well-defined problem space, yet that system still required the conceptual primitives of physical chemistry and evolutionary co-evolutionary constraints that were established by prior human science. For paradigm-shifting innovation, however, the constrained-manifold pattern suggests that the required conceptual moves—recognising framework inadequacy, proposing new primitives, evaluating whether new frameworks are coherent—are not achievable by current architectures without external conceptual input.

Faggin’s framework [36] provides one theoretical lens. Faggin argues that genuine understanding requires phenomenological access: first-person awareness of what symbols mean, not only the ability to manipulate them via statistical regularities. This perspective is relevant here because it articulates why systems trained on distributional co-occurrence may face a principled barrier to ontological novelty: without semantic self-access, a model may be unable to detect that its current primitive ontology is insufficient. Whether this reflects the absence of consciousness in current LLMs or a narrower architectural limitation remains an open empirical question.

5.5 Planned Expert Validation Study

To address external validity of automated labels, future work will run a stratified expert annotation study on 30 proposals (5 sampled from each major structural

grouping). Three annotators are planned: one sensory biologist, one philosopher of perception, and one robotics researcher. Each proposal will receive a binary novelty judgment plus free-text justification. Inter-rater reliability will be quantified with Krippendorff’s α . Disagreements between expert labels and automated classification will be analyzed as directional bias (false-novel vs. false-traceable) and used to recalibrate thresholding and decomposition rules.

5.6 Limitations

Several methodological limitations constrain the conclusions.

Single domain. The study uses one conceptual domain (sensory modalities). Findings may not generalise to conceptual domains with different structural properties, such as mathematical objects, physical theories, or social organisational forms.

Automated evaluation. All evaluation methods are automated. Conservative thresholds mean we likely underestimate AI creativity, but the methods cannot capture all nuances an expert might perceive. Future work should include expert validation of a subset of proposals to calibrate and improve automated methods.

Corpus scope. Traceability is currently defined relative to academic mainstream literature in sensory biology, robotics sensing, and computational neuroscience. Non-academic sources (e.g., science-fiction corpora, patent databases, and broader philosophical text collections) were not included and may contain additional apparent predecessors. In addition, OpenAlex/Crossref coverage and metadata quality are heterogeneous across venues and years (including abstract availability and indexing lag), which can bias corpus composition and therefore observed traceability rates.

Prompt robustness. Full prompt-paraphrase grids were not included in the canonical experimental block. Conclusions should be read as structural tendencies under the present protocol, not as invariances proven over every prompting regime.

Temperature range. Temperature ablations were not systematically applied; runs were conducted at $T = 0.7$. Extending to broader temperature ranges and verifying that the null result persists would strengthen confidence.

Generative architecture coverage. All tested models are transformer-based autoregressive LLMs. The No New Basis Theorem applies to a wider class of gradient-descent learners, but empirical coverage of non-autoregressive or hybrid architectures is absent.

Interpretation of the No New Basis Theorem. The theorem provides formal motivation for the constrained-manifold interpretation but rests on assumptions (fixed vocabulary, continuous gradient-based learning) that may not hold for future architectures incorporating discrete search, external memory, or online learning components.

6 Conclusions

Current large language models demonstrate remarkable facility with lexical and structural creativity—recombining, extending, and remixing known conceptual elements in novel configurations. Yet this study reveals a systematic boundary: ontological innovation—the introduction of fundamentally new primitives—remains uniformly

absent. Across seven state-of-the-art models and 168 generated theories, zero proposals achieved confirmed framework-external novelty. This finding aligns with formal constraints from the No New Basis Theorem and suggests that current transformer architectures remain bounded by their training-derived representational manifold. The traceability threshold should be interpreted as an operational indicator for calibrated inference, not as an absolute classifier of whether an output is intrinsically traceable. For scientific practice, the implication is clear: AI systems can exhaustively explore and articulate known theoretical space, but should not be positioned as autonomous sources of paradigm-level innovation without human conceptual intervention.

Declarations

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Competing Interests

The authors declare no competing interests.

Ethics Approval

Not applicable.

Author Contributions

All authors contributed to design, analysis, and writing.

Data and Code Availability

Data, prompts, model responses, and analysis code are publicly available at https://github.com/JABE22/PhD/tree/main/research/test1_ontological-innovation.

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